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Aim of The Experiment

Investigate logic behaviour of AND, OR, NOT, NAND, NOR, Ex-OR, Ex-NOR gates.

Objectives

To design and verify the truth table of different logic gate

Apparatus Required:

SL NO	Name of Component Required	Specification	Quantity
1	Trainer kit (Analog digital kit)	XDO-ANADIGI	1
2.	Bread board	–	1
3.	AND Gate (IC-74LS08)	Quad 2 input AND Gate	1
4.	OR Gate (IC-74LS 32)	Quad 2 input OR Gate	1
5	NOT Gate (IC-74LS 04)	Hex inverter	1
6	NAND Gate (IC-74LS 00)	Quad 2 input NAND Gate	1
8	Ex-OR Gate (IC-74LS86)	Quad 2 input XOR Gate	1
9.	NOR Gate (IC 74LS02)	Quad 2 input NOR Gate	1
10.	Connecting wires	–	As per requirement

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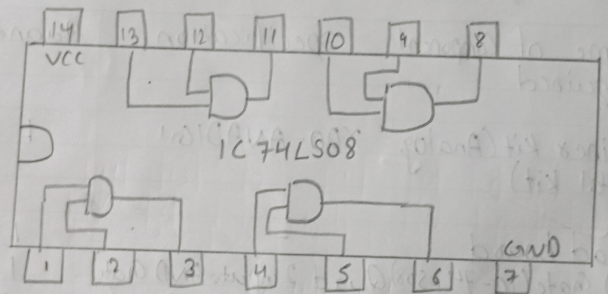

Fig 1.1: AND Gate IC 74LS08 Pin Diagram

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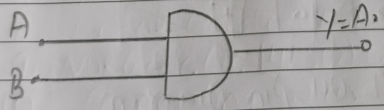
Theory:

AND, OR, NOT Gate are basic gate. Ex-OR and Ex-NOR are special gates. NAND and NOR are universal gates. Basically, logic gates are electronic circuits because they are made up of no of electronic device and components. Inputs and Outputs of logic gates can occur in 2 levels. The 2 level are term high and low or True and False or ON and OFF, or simply 1 and 0

AND Gate

An AND Gate has 2 or more input but only one output. The output assume logic 1 state only when each of its input is at logic 1 state. The output assums logic 0 ~~states~~ even if one of its input is at logic 0 state AND Gates is also known as "on all or nothing" Gate.

Logic Symbol:-



Truth Table

A	B	Y = A.B
0	0	0
0	1	0
1	0	0
1	1	1

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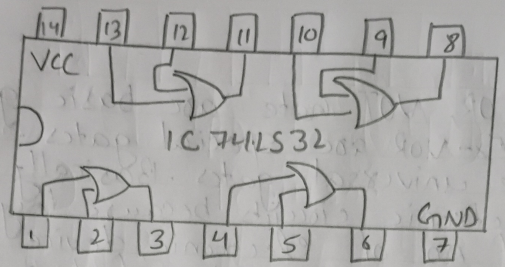

Fig 1.2: OR Gate IC 74LS32 pin Diagram

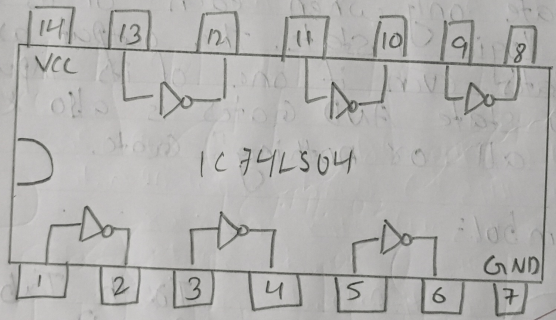

Fig 1.3: NOT Gate IC 74LS04 pin Diagram

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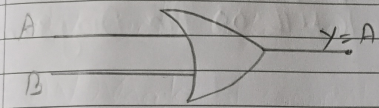
OR Gate :-

An OR Gate may have 2 or more inputs but only 1 output. The output assumes the logic 1 state, even if one of its input is in logic 1 state. OR gate is also "any or all" gate.

Logic Symbol



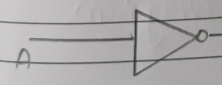
Truth Table

A	B	Y=A+B
0	0	0
0	1	1
1	0	1
1	1	1

NOT Gate :

A NOT Gate is also known as inverter, has only 1 input and only 1 output. It is a device whose output is always the complement of its input. That is the output of a NOT gate assume the Logic 1 gate state its input is in logic 0 state and vice versa

Logic Symbol



Truth Table

A	Y=Ā
0	1
1	0

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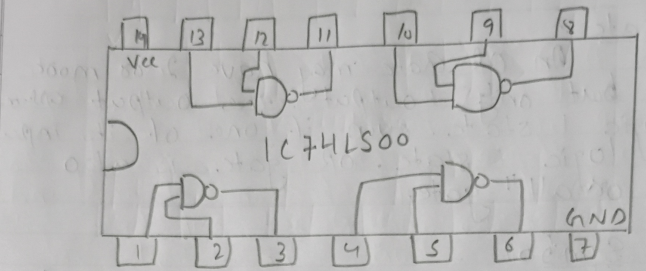

Fig 1.4: NAND Gate IC74LS00 Pin Diagram

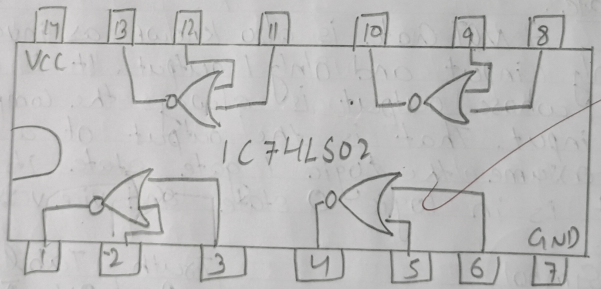

Fig 1.5 : NOR Gate IC74LS02 Pin Diagram

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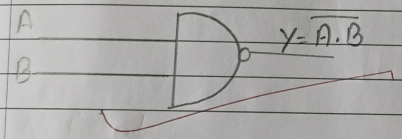
NAND Gate

A NAND Gate is an universal gate. It can perform all basic logic function. NAND mean NOT gate with AND Gate. The output is logic 0 level, only when each of its input assume a logic level 1 only when each of its input assumes a logic 1 level. for any other combination of inputs the output is logic is 1 level

Logic Symbol



Truth Table

A	B	Y = $\overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

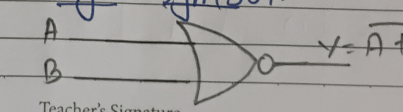
NOR Gate

A NOR Gate is an universal Gate. It can perform all the basic logic function. NOR means NOT Gate with OR Gate. The output is logic 1 level only when each of its input assumes a logic 0 level. For any other combination of input, the output is logic 0 level.

Truth Table

A	B	Y = $\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

Logic Symbol:-



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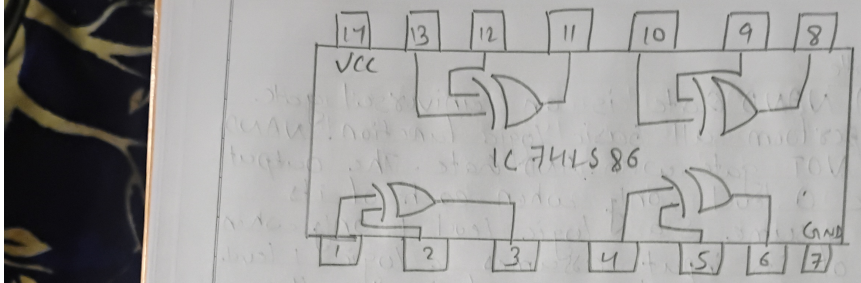

Fig 1.6 : X-OR Gate Ic 74LS 86 Pin Diagram

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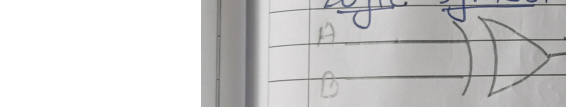
Ex-OR Gate

An Ex-OR Gate is a two input, one output logic circuit, whose output assumes a logic 1 state when 1 and 1 of its 2 input assumes a Logic state. Under the condition when both the inputs are same either 0 or 1 the output assumes a Logic 0.

Logic Symbol



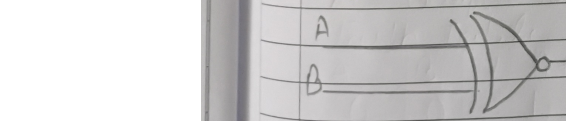
Truth Table

A	B	Y = A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

Ex-NOR Gate

An Ex-NOR Gate is an 2 input, 1 output logic circuit whose output assumes a logic 1 state when 2 input are same and it assume a logic state 0 when ony one of its input assumes a logic state 1.

Logic Symbol:



Truth Table

A	B	Y = A⊕B
0	0	1
0	1	0
1	0	0
1	1	1

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PROCEDURE :-

1. The ICs corresponding to the NOT, OR, AND NAND, NOR, X-OR gate were placed properly on the bread board.
2. The power supply was connected to attaching pin no 14. 7 to ground and pin no 14 to +5V using patch chords
3. The input pins of each ICs were connected to the corresponding logic input with patch chord as per the configuration on diagram provided
4. The output pin of each ICs were connected to LED indicate to observe the logic output.
5. The power supply of the bread board was switched on and combination of logic input were applied to the input pins of each IC.
6. The corresponding output were observed and recorded to varify the truth table of each gate.
7. The power supply was switched off after completing the observation for each IC

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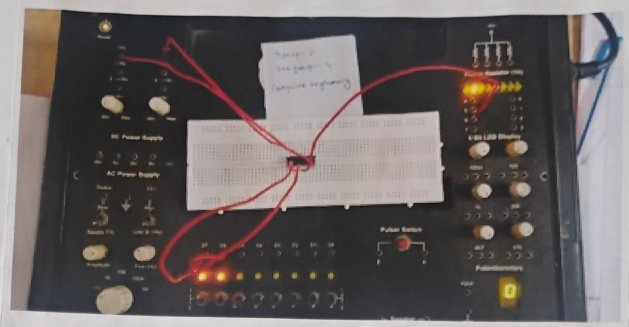
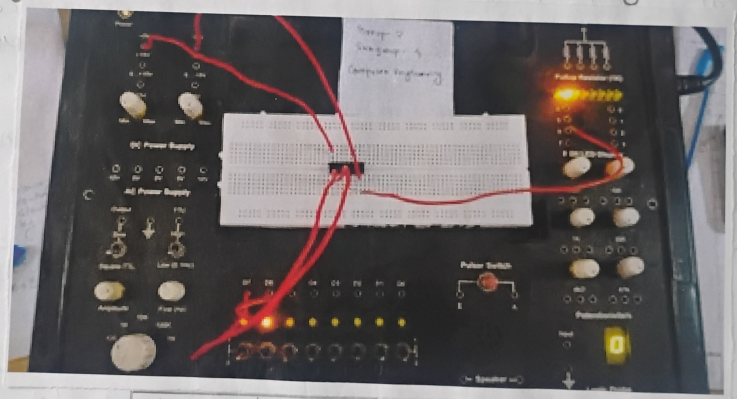


Input		Output
A (D_7)	B (D_6)	Y = A·B (D_5)
ON	ON	Glowing

ON → 1
OFF → 0

Fig 1.7: AND Gate IC74LS08 Circuit diagram



Input		Output
A(D_7)	B(D_6)	Y = A+B (D_5)
OFF	ON	Glowing

ON → 1
OFF → 0

Fig 1.8: OR Gate IC74LS32 Circuit Diagram

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8. The steps were repeated for all IC. to ensure that the logic behaviour of NOT OR, AND, NAND, NOR, Ex-OR, Ex-NOR gate were throughly invested and their truth table observed.

Observation Table:-

1) AND GATE:

Observation Table 1:-

Input		Output	
A (D_7)	B (D_6)	$Y = A \cdot B$ (G)	Remark
OFF	OFF	Not Glowing	
OFF	ON	Not Glowing	pratical output is matching with theoritical output
ON	OFF	Not Glowing	
ON	ON	Glowing	

OFF → 0 , ON → 1

2. OR Gate:

Observation Table 2:-

Input		Output	
A (D_7)	B (D_6)	$Y = A+B$ (G)	Remark
OFF	OFF	Not Glowing	
OFF	ON	Glowing	pratical output is matching Theoritical output
ON	OFF	Glowing	
ON	ON	Glowing	

ON → 1 , OFF → 0

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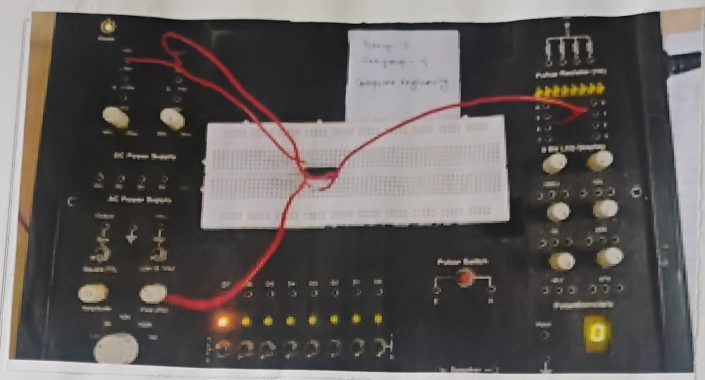
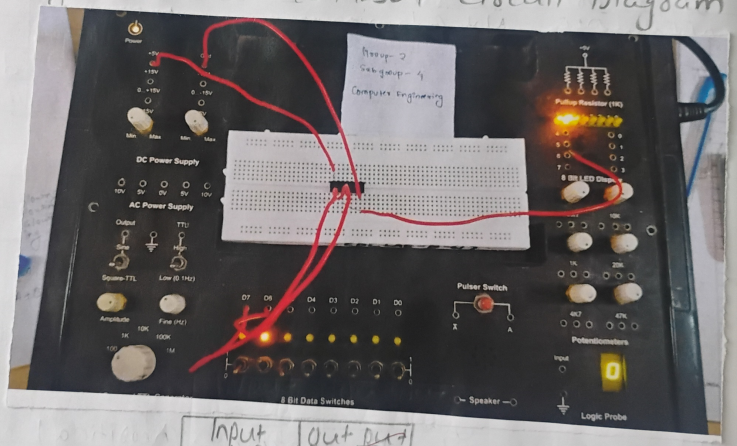


Input	Output
A (0)	Y = $\overline{A}$ (1)
ON	Glowing

ON → 1

Fig 1.9: NOT Gate IC 74LS04 Circuit Diagram



Input A (D7)	Input B (D6)	Output Y = $\overline{A \cdot B}$ (D1)
OFF	ON	Glowing

OFF → 0
ON → 1

Fig 1.10: NAND Gate IC 74LS00 Circuit Diagram

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5.

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3. NOT GATE

Observation Table 3

Input A (D_7)	Output $Y=\overline{A}$ (6)	Remark
ON	Not Glowing	Pratical output is matching with theoritical output
OFF	Glowing	

ON → 1, OFF → 0,

4. NAND GATE :-

Observation Table 4

Input A (D_7)	Input B (D_6)	output $Y=\overline{A.B}$ (6)	Remark
OFF	OFF	Glowing	pratical output is matching with theoritical output
OFF	ON	Glowing	
ON	OFF	Glowing	
ON	ON	Not Glowing	

ON → 1, OFF → 0,

5. NOR Gate:

Observation Table 5

Input A (D_7)	Input B (D_6)	Output $Y=\overline{A+B}$ (7)	Remark
OFF	OFF	Glowing	pratical Output is matching with theoritical output
OFF	ON	Not Glowing	
ON	OFF	Not Glowing	
ON	ON	Not Glowing	

ON → 1, OFF → 0

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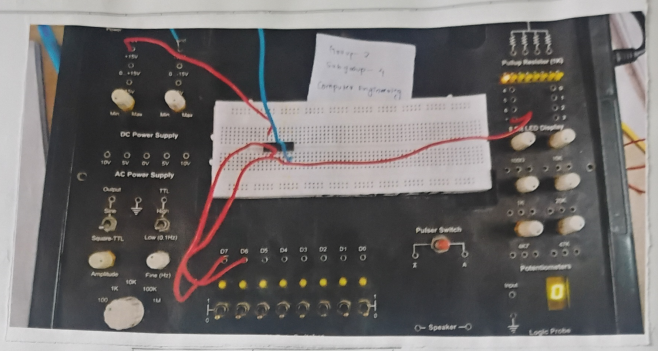
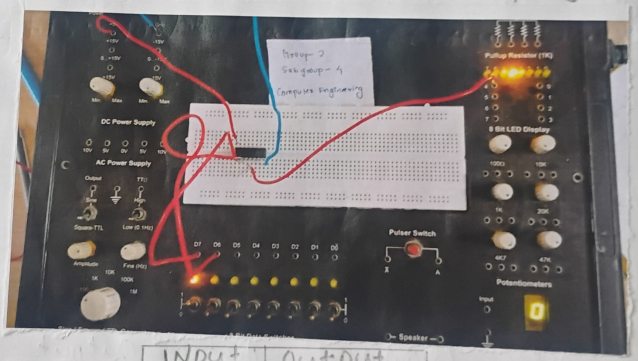


Input		Output
A	B	Y = A+B
(D7)	(D6)	(7)
OFF	OFF	Glowing

OFF → 0

Fig 1.11: NOR Gate IC74LS02 Circuit Diagram



INput		Output
A	B	Y = A(+)B
D7	(D6)	(1)
ON	OFF	Glowing

ON → 1
OFF → 0

Fig 1.12: X-OR Gate IC74LS86 Circuit Diagram

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6. Ex-OR Gate

Observation Table 6

Input A (D_7)	Input B (D_6)	Output Y = A⊕B (4)	Remark
OFF	OFF	Not Glowing	pratical output is matching with the theoritical output
OFF	ON	Glowing	
ON	OFF	Glowing	
ON	ON	Not Glowing	

ON → 1, OFF → 0

7. Ex-NOR Gate

Observation Table 7

Input A (D_7)	Input B (D_6)	Output Y = A⊕B (6)	Remark
OFF	OFF	Glowing	pratical output is matching with the theoritical output
OFF	ON	Not Glowing	
ON	OFF	Not Glowing	
ON	ON	Glowing	

ON → 1, OFF → 0

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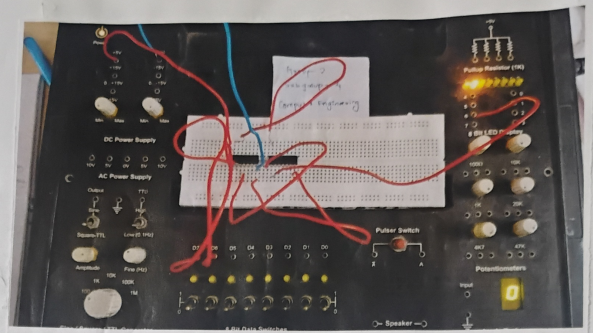

Input		Output
A (D7)	B (D6)	$Y = \overline{A \oplus B}$ (6)
OFF	OFF	Glowing

OFF → 0

Fig 1.13 : X-NOR Gate IC Circuit Diagram

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Conclusion

The logic behaviour of different types of logic gates have been successfully investigated and verified. The outputs obtained for AND, OR, NOT, NAND, XOR and X-NOR gates are found to be in aggrement with their respective truth tables. Thus the basic operation and functionality of different logic gates have been studied successfully.

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